The channel-irreducible vertex of the Hubbard atom away from half filling: derivation

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1 Overview

The irreducible vertex Γ_r of the Hubbard atom is known in closed form at half filling [1]. Here we derive it at arbitrary filling (zero magnetic field) in all four channels $r \in \{\mathrm{d}, \mathrm{m}, \mathrm{s}, \mathrm{t}\}$. By definition $\Gamma_r = \beta^2(\chi_r^{-1} - \eta_r \chi_{0,r}^{-1})$, where χ_r is the generalized susceptibility, an infinite matrix in two fermionic Matsubara frequencies. So the task is to invert χ_r in closed form.

At half filling χ_r is a diagonal matrix plus a mirror term plus a matrix of rank at most two, and the inversion is elementary. Away from half filling the mirror term becomes a dense *Cauchy matrix* $C_{nm} = \rho/(z_n + z_m)$ (Figure 1). The particle–hole susceptibility is then no longer a finite-rank perturbation of an easily invertible matrix. Two properties of C still allow a closed-form inverse:

1. $C^2 = h^2 \mathbb{1}$ for a number h: like the mirror it replaces, C is a scaled involution.
2. C maps a simple pole to a simple pole: $\sum_m C_{nm}/(z_m - x) = \alpha(x)/(z_n + x)$, with a thermal factor obeying $\alpha(x)\alpha(-x) = h^2$.

After multiplication by a polynomial, χ_r is a diagonal matrix plus λC plus a matrix of rank two. Each column of χ_r^{-1} is then built from a diagonal entry, one Cauchy column, a constant, and two pole pairs with one thermal denominator each. All coefficients follow from 2×2 linear systems whose determinants are given by a single quartic.

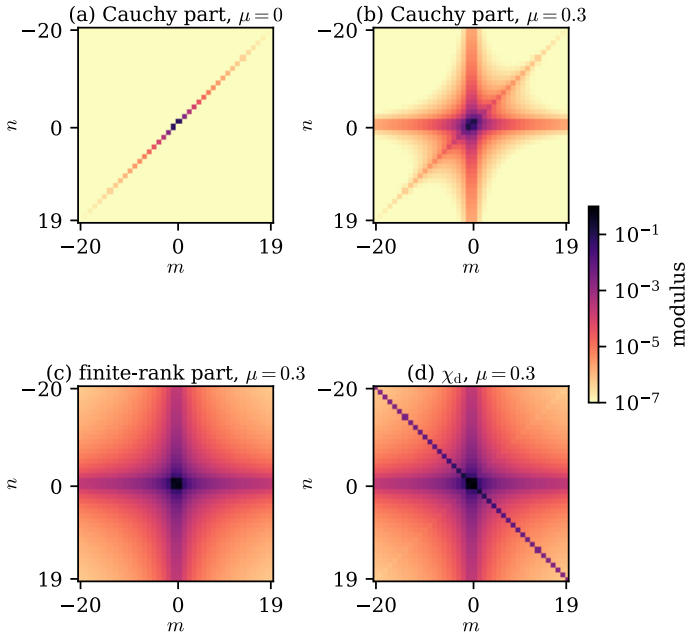


Figure 1: Structure of χ_d at $U = 2$, $\beta = 3$, $\omega = 0$, as moduli of matrix elements in the fermionic indices n, m. (a) At half filling the pair term is a mirror, nonzero only for $m = -n - 1$. (b) Away from half filling it becomes the dense Cauchy matrix C. (c) The remaining finite-rank part. (d) The full χ_d; the diagonal dominates.

Results.

- **Triplet:** Γ_t is a diagonal-minus-mirror term (Theorem 2).

- **Singlet:** Γ_{s} is a diagonal-plus-mirror term, the constant $2U$, and one separable term with a single thermal denominator (Theorem 3).
- **Density and magnetic:** $\Gamma_{\mathrm{d}}, \Gamma_{\mathrm{m}}$ have the same anatomy: a diagonal term, the constant $\pm U$, a dense term without thermal factors, and two separable terms with one thermal denominator each (Theorem 4). At half filling the dense term becomes a mirror term and one separable term remains (Corollary 5).

The inverses are two-sided and agree with the limit of inverses on finite frequency boxes, i.e. with what numerical inversion converges to (Proposition 6).

Roadmap.
- Section 2 defines all quantities; Section 3 states the one external input, the susceptibility from the Lehmann representation.
- Section 4 states all results. The rest of the note proves them.
- Section 5 evaluates the Matsubara sums; Section 6 proves the particle–particle results, which need no Cauchy matrix.
- Section 7 develops the algebra of C, which Section 8 uses for the particle–hole channels. The limit $\mu \to 0$ and the finite-box statement are proved in Appendix B.

What is assumed. Everything is proved from the input Eqs. 13–15 (Ref. [2], checked against exact diagonalization) and the rational identities of Appendix A (checked by computer algebra). Independent numerical checks are listed in Appendix D. Numerical stability in floating point is not discussed here.

Implementation. The results are implemented as `DopedHubbardAtom` in the Julia package HubbardAtoms.jl, which also evaluates them stably at low temperatures and near the removable singularities, and is tested against exact diagonalization and finite-box inversion.

2 Setup

2.1 Model and thermal weights

$$\mathcal{H} = U\Big(n_\uparrow - \frac{1}{2}\Big)\Big(n_\downarrow - \frac{1}{2}\Big) - \mu(n_\uparrow + n_\downarrow - 1), \quad a \coloneqq \frac{U}{2}, \tag{1}$$

with real U, μ (the chemical potential measured from half filling, $\mu = 0$) and inverse temperature $\beta > 0$. The four states $|0\rangle, |\uparrow\rangle, |\downarrow\rangle, |\uparrow\downarrow\rangle$ have energies $\frac{U}{4} + \mu$, $-\frac{U}{4}$, $-\frac{U}{4}$, $\frac{U}{4} - \mu$ and Boltzmann weights p_0, p_1, p_1, p_2 (p_1 is the weight of *each* singly occupied state). Hence

$$p_0 + 2p_1 + p_2 = 1, \qquad \frac{p_2}{p_0} = e^{2\beta\mu}, \qquad \frac{p_1}{p_0} = e^{\beta(a+\mu)}, \qquad \frac{p_1}{p_2} = e^{\beta(a-\mu)}. \tag{2}$$

Define

$$\rho \coloneqq p_2 - p_0 = \langle n_\uparrow + n_\downarrow \rangle - 1, \qquad h \coloneqq \beta\sqrt{p_0 p_2}, \qquad \overline{c} \coloneqq \beta(p_0 + p_2)/2. \tag{3}$$

From Eq. 2, $p_1 + p_2 = (1+\rho)/2$, $p_0 + p_1 = (1-\rho)/2$, and

$$\rho^2 + 4p_0p_2 = (p_0 + p_2)^2, \qquad \overline{c}^2 - h^2 = \beta^2\rho^2/4. \tag{4}$$

For $\mu \neq 0$, inserting $p_2 = p_0 e^{2\beta\mu}$:

$$\begin{aligned} &\rho = 2p_0 e^{\beta\mu}\sinh(\beta\mu), \quad p_0 + p_2 = 2p_0 e^{\beta\mu}\cosh(\beta\mu) \\ \Longrightarrow \quad &\rho\coth(\beta\mu) = p_0 + p_2, \quad \rho^2 = 4p_0p_2\sinh^2(\beta\mu). \end{aligned} \tag{5}$$

2.2 Frequencies and propagator

Fermionic and bosonic Matsubara frequencies are $\nu_n = (2n+1)\pi/\beta$ and $\Omega_\ell = 2\ell\pi/\beta$ ($n, \ell \in \mathbb{Z}$). The bosonic transfer $\omega_k \coloneqq \Omega_k$ is fixed throughout. Write

$$x_n \coloneqq \mu + i\nu_n, \qquad w \coloneqq i\omega_k, \qquad J(x) \coloneqq \frac{1}{x^2 - a^2}. \tag{6}$$

The propagator $G(\nu) \coloneqq -\int_0^\beta \mathrm{d}\tau\, e^{i\nu\tau}\langle T c_\uparrow(\tau) c_\uparrow^\dagger\rangle$ follows from the Lehmann representation: $c_\uparrow^\dagger$ maps $|0\rangle \to |\uparrow\rangle$ (energy change $-a-\mu$, weight p_0+p_1) and $|\downarrow\rangle \to |\uparrow\downarrow\rangle$ (energy change $a-\mu$, weight p_1+p_2). So

$$G(\nu_n) = \frac{p_0+p_1}{x_n + a} + \frac{p_1+p_2}{x_n - a} = G(x_n) \coloneqq \frac{x_n + a\rho}{x_n^2 - a^2}, \tag{7}$$

where we use the same letter for the rational function $G(x)$.

2.3 Susceptibilities and the irreducible vertex

Following Ref. [1], the generalized susceptibility is

$$\begin{aligned}\chi_{\sigma\sigma'}(\nu,\nu',\omega) \coloneqq \iiint_0^\beta \mathrm{d}\tau_1\, \mathrm{d}\tau_2\, \mathrm{d}\tau_3\, e^{-i\nu\tau_1 + i(\nu+\omega)\tau_2 - i(\nu'+\omega)\tau_3} \\ \times \Big[\langle T c_\sigma^\dagger(\tau_1) c_\sigma(\tau_2) c_{\sigma'}^\dagger(\tau_3) c_{\sigma'}(0)\rangle - \langle T c_\sigma^\dagger(\tau_1) c_\sigma(\tau_2)\rangle\langle T c_{\sigma'}^\dagger(\tau_3) c_{\sigma'}(0)\rangle\Big].\end{aligned} \tag{8}$$

Its particle–particle form is $\chi_{\mathrm{pp},\sigma\sigma'}(\nu,\nu',\omega) \coloneqq \chi_{\sigma\sigma'}(\nu,\nu',-\omega-\nu-\nu')$. The channels and bare susceptibilities are

$$\chi_\mathrm{d} = \chi_{\uparrow\uparrow} + \chi_{\uparrow\downarrow}, \quad \chi_\mathrm{m} = \chi_{\uparrow\uparrow} - \chi_{\uparrow\downarrow}, \quad \chi_\mathrm{s} = \frac{-\chi_{\mathrm{pp},\uparrow\uparrow} + 2\chi_{\mathrm{pp},\uparrow\downarrow} - 2\chi_{0,\,\mathrm{pp}}}{4}, \quad \chi_\mathrm{t} = \frac{\chi_{\mathrm{pp},\uparrow\uparrow} + 2\chi_{0,\,\mathrm{pp}}}{4}, \tag{9}$$

$$\chi_{0,r}(\nu,\nu',\omega) = \delta_{\nu\nu'} B(\nu), \qquad B(\nu) = \begin{cases} -\beta G(\nu) G(\nu+\omega) & (r = \mathrm{d},\mathrm{m}), \\ -\frac{\beta}{2} G(\nu) G(-\nu-\omega) & (r = \mathrm{s},\mathrm{t}), \end{cases} \tag{10}$$

with $\chi_{0,\,\mathrm{pp}} \coloneqq \chi_{0,\,\mathrm{s}}$. At fixed ω_k all of these are matrices indexed by (n,m) with $\nu = \nu_n$, $\nu' = \nu_m$; we write $B_n \coloneqq B(\nu_n)$ and $\boldsymbol{B} \coloneqq \mathrm{diag}(B_n)$. The channel-irreducible vertex is

$$\Gamma_r = \beta^2(\chi_r^{-1} - \eta_r \boldsymbol{B}^{-1}), \qquad \eta_\mathrm{d} = \eta_\mathrm{m} = \eta_\mathrm{t} = 1, \quad \eta_\mathrm{s} = -1. \tag{11}$$

2.4 Lattice and mirror

Define the centered lattice and the mirror index

$$\begin{gathered} z_n \coloneqq x_n + \frac{w}{2} = \mu + i\pi\frac{2n+1+k}{\beta}, \qquad \overline{n} \coloneqq -n-k-1, \\ z_{\overline{n}} = 2\mu - z_n, \qquad x_{\overline{n}} = 2\mu - w - x_n. \end{gathered} \tag{12}$$

The lattice lies on the line $\mathrm{Re}\, z = \mu$, and the mirror reflects it about the real axis (Figure 2). The mirror matrix is $\mathcal{M}_{nm} \coloneqq \delta_{m\overline{n}}$, with $\mathcal{M}^2 = \mathbb{1}$. Products of infinite matrices are taken entrywise. A sum $\sum_n$ is the limit of partial sums over the mirror-symmetric boxes $\{n : |2n+1+k| \le 2M+1\}$, $M \to \infty$; we write PV ("principal value") where it converges only in this sense. All transposes are ordinary, not Hermitian. Symbols introduced inside a proof are local to it. A function φ on $\mathbb{C}$ is *proper rational* if it is rational and $\varphi(t) \to 0$ for $t \to \infty$.

3 Input: the generalized susceptibility

The Lehmann representation of $\chi_{\sigma\sigma'}$ for the atom is evaluated in Ref. [2] [Eq. (D3)]. At zero field and in our notation it reads

$$\chi_{\uparrow\uparrow,nm} = -\beta G(x_n) G(x_n + w)\delta_{nm} + \beta a^2(1-\rho^2) J(x_n + w) J(x_m)(\delta_{k0} - \delta_{nm}), \tag{13}$$

$$\chi_{\uparrow\downarrow,nm} = -4\beta a^2 p_1 J(x_n) J(x_n + w)\delta_{nm} + 4\beta a^2(p_0 p_2 - p_1^2) J(x_n) J(x_m)\delta_{k0} + \mathcal{N}_{nm}, \tag{14}$$

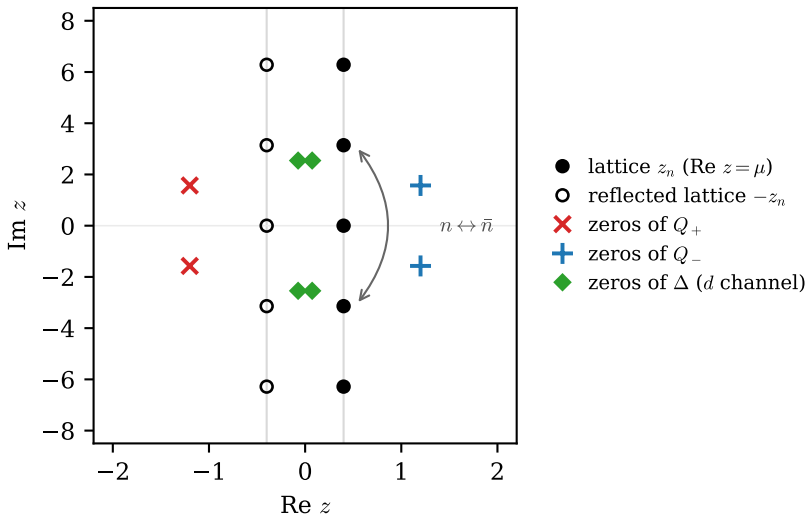


Figure 2: The complex plane for $U = 2.4$, $\beta = 2$, $\mu = 0.4$, $\omega = \omega_1$. The lattice z_n (filled) lies on Re $z = \mu$; the reflected lattice $-z_n$ (open), where the Cauchy kernel $1/(z + z_m)$ is singular, lies on Re $z = -\mu$. Crosses: zeros of $Q_\pm$ (Section 3); diamonds: zeros of e (Section 4.2). None lies on the lattice, so all lattice sums below reduce to finitely many residues.

$$\mathcal{N}_{nm} := c_{n+m+k+1}\Pi(x_n, x_m) + R_{\text{reg}}(x_n, x_m), \qquad c_\ell := \frac{\rho}{2\mu + i\Omega_\ell},$$
$$\Pi(x, y) := \left(\frac{1}{x + w - a} + \frac{1}{y - a}\right)\left(\frac{1}{y + w - a} + \frac{1}{x - a}\right). \tag{15}$$

The pair coefficient c_ℓ stems from the transition $|0\rangle \leftrightarrow |\uparrow\downarrow\rangle$. At $\mu = 0$ it is understood as its limit, $c_\ell = \beta p_0 \delta_{\ell 0}$ (by Eq. 5, $\rho/2\mu \to \beta p_0$). The regular part R_{reg} is a sum of twelve simple fractions, reproduced in Appendix C. We take Eqs. 13–15 as given; they agree with an independent exact diagonalization of the atom to 10^{-13} (Appendix D).

Compact form. Define the **Cauchy matrix** $C_{nm} := c_{n+m+k+1}$. Since $2\mu + i\Omega_{n+m+k+1} = z_n + z_m$ by Eq. 12,

$$C_{nm} = \frac{\rho}{z_n + z_m} \quad (\mu \neq 0), \qquad C_{nm} = \beta p_0 \delta_{m\bar{n}} \quad (\mu = 0), \tag{16}$$

and $z_n + z_m = 2\mu + i(\cdots) \neq 0$ for $\mu \neq 0$. Further let

$$Q_\pm(z) := (z \pm a)^2 - \frac{w^2}{4}, \qquad q_0 := a^2 - \frac{w^2}{4} = (U^2 + \omega_k^2)/4. \tag{17}$$

Since $x_n = z_n - w/2$,

$$(x_n - a)(x_n + w - a) = Q_-(z_n), \qquad (x_n + a)(x_n + w + a) = Q_+(z_n), \qquad Q_\pm(-z) = Q_\mp(z). \tag{18}$$

Neither $Q_\pm(z_n)$ vanishes: its factors have imaginary parts ν_n and $\nu_n + \omega_k$, both nonzero.

Lemma 1 (compact input). With $z = z_n$ and $v = z_m$,

$$\mathcal{N}_{nm} = \frac{4a^2 C_{nm} - 2a}{Q_-(z)Q_-(v)} + \frac{4a^2(1 - \rho)(z + v + 2a)(zv + q_0)}{Q_+(z)Q_-(z)Q_+(v)Q_-(v)}. \tag{19}$$

Proof. Put $y := z + v = x_n + x_m + w$. The two factors of Π combine to

$$\Pi(x_n, x_m) = \frac{y - 2a}{(x_n + w - a)(x_m - a)} \cdot \frac{y - 2a}{(x_m + w - a)(x_n - a)} = \frac{(y - 2a)^2}{Q_-(z)Q_-(v)} \tag{20}$$

by Eq. 18. For $\mu \neq 0$, $(y - 2a)^2/y = y - 4a + 4a^2/y$, so $C_{nm}(y - 2a)^2 = \rho(y - 4a) + 4a^2 C_{nm}$. For $\mu = 0$ this holds too, because $\rho = 0$ and $C_{nm} \neq 0$ only where $y = 0$. Thus Eq. 19 is equivalent to

$$R_{\text{reg}}\Big(z - \frac{w}{2}, v - \frac{w}{2}\Big) = \frac{-2a - \rho(z + v - 4a)}{Q_-(z)Q_-(v)} + \frac{4a^2(1 - \rho)(z + v + 2a)(zv + q_0)}{Q_+(z)Q_-(z)Q_+(v)Q_-(v)}, \tag{21}$$

which is the rational identity (A1) of Appendix A. □

4 Results

All lattice sums that occur below are expressed through

$$S_k(x) \coloneqq \begin{cases} -\frac{\beta}{2}\tanh(\beta(x - \mu)/2) & k \text{ even} \\ -\frac{\beta}{2}\coth(\beta(x - \mu)/2) & k \text{ odd} \end{cases}, \tag{22}$$

which equals $\mathrm{PV}\sum_n (z_n - x)^{-1}$ for x off the lattice (Lemma 7).

4.1 Singlet and triplet

Let $b \coloneqq 2\mu - w$ and $t_n \coloneqq x_n(b - x_n)$; by Eq. 12, $x_{\overline{n}} = b - x_n$, so $t_{\overline{n}} = t_n$. Define

$$\mathcal{P}(t) \coloneqq (t + a^2)^2 - a^2 b^2, \qquad \mathcal{G}(t) \coloneqq t + a\rho b + a^2\rho^2, \qquad A_{\text{s}} \coloneqq a\rho b + a^2(1 - 8p_1), \tag{23}$$

$$\mathcal{T}_k(t) \coloneqq -\frac{S_k(\mu + \xi)}{\xi}, \qquad \xi \coloneqq \sqrt{b^2/4 - t}. \tag{24}$$

$\mathcal{T}_k$ is even in ξ, so either branch may be used (for $\xi = 0$ and even k, take the limit $\beta^2/4$). It equals $\sum_n (t_n - t)^{-1}$ (Lemma 9).

Theorem 2 (triplet). For $n \neq \overline{n}$, assuming $t_n + a\rho b + a^2 \neq 0$,

$$\Gamma_{\text{t},nm} = \frac{\beta a^2(1 - \rho^2)\mathcal{P}(t_n)}{\mathcal{G}(t_n)(t_n + a\rho b + a^2)}(\delta_{nm} - \delta_{m\overline{n}}). \tag{25}$$

At a fixed point $n = \overline{n}$ (odd k, $n = -(k + 1)/2$), $\Gamma_{\text{t},nm} = \Gamma_{\text{t},mn} = 0$ for all m.

Theorem 3 (singlet). Assume

$$K_{\text{s}} \coloneqq 1 + \frac{2a}{\beta}\mathcal{T}_k(-A_{\text{s}}) \neq 0. \tag{26}$$

Then

$$\Gamma_{\text{s},nm} = \frac{\beta a^2(\rho^2 - 1 + 8p_1)\mathcal{P}(t_n)}{\mathcal{G}(t_n)(t_n + A_{\text{s}})}(\delta_{nm} + \delta_{m\overline{n}}) + 4a - \frac{4a\,\mathcal{P}(-A_{\text{s}})}{K_{\text{s}}(t_n + A_{\text{s}})(t_m + A_{\text{s}})}. \tag{27}$$

If $t_n + A_{\text{s}} = 0$ for some n, the entries in the rows and columns n and $\overline{n}$ are the limits of Eq. 27 as μ approaches its value; the divergent parts cancel.

At $\mu = 0$ both formulas reduce to Eq. 19 of Ref. [1] (Appendix D).

4.2 Density and magnetic

Let $\varepsilon \coloneqq +1$ for $r = \text{d}$ and $\varepsilon \coloneqq -1$ for $r = \text{m}$. Define

$$H(z) \coloneqq z^2 + 2a\rho z + q_1, \qquad q_1 \coloneqq q_0 + 4\varepsilon a^2 p_1, \qquad \lambda \coloneqq 4\varepsilon a^2,$$
$$P(z) \coloneqq Q_+(z)Q_-(z), \qquad G_2(z) \coloneqq (z + a\rho)^2 - \frac{w^2}{4}, \tag{28}$$

$$e(z) \coloneqq H(z)H(-z) - \lambda^2 h^2/\beta^2 = \left(z^2 + q_1\right)^2 - 4a^2\rho^2 z^2 - 16a^4 p_0 p_2 = (z^2 - s_1)(z^2 - s_2), \tag{29}$$

where $s_{1,2}$ are the roots of $s^2 + 2(q_1 - 2a^2\rho^2)s + q_1^2 - 16a^4 p_0 p_2$, and u_j is either square root of s_j. For $r = \mathrm{m}$ and $k = 0$, $q_1 = a^2(1 - 4p_1)$, so $H(a) = 4a^2 p_2$ and $H(-a) = 4a^2 p_0$ by Eq. 2. Then $e(a) = 0$: one root is $s_1 = a^2$, at the double zeros $\pm a$ of P. We call this root pair *atomic*. With the thermal function

$$\alpha_k(x) \coloneqq \rho\, S_k(x) - \overline{c}, \tag{30}$$

define

$$F(z) \coloneqq \frac{P(z)}{e(z)}, \qquad g_j(z) \coloneqq \frac{1}{z^2 - s_j},$$
$$\mathbb{D}(z, v) \coloneqq \frac{zF(z) - vF(v)}{z^2 - v^2}, \qquad \mathbb{D}(z, z) \coloneqq \frac{F(z) + zF'(z)}{2z}, \tag{31}$$

$$K_j \coloneqq \beta H(-u_j) + \lambda\alpha_k(u_j), \qquad Z_j \coloneqq \frac{u_j P(u_j)\left[\beta H(-u_j) - \lambda\alpha_k(u_j)\right]}{K_j}, \qquad \gamma_j \coloneqq \frac{\lambda\rho Z_j}{s_j - s_\ell}, \tag{32}$$

with $\ell \neq j$.

Theorem 4 (density and magnetic). Let $\mu \neq 0$, $e(z_n) \neq 0$ for all n, and $K_j \neq 0$. Then

$$\Gamma_{r,nm} = \beta P(z_n)\left[\frac{1}{G_2(z_n)} - \frac{H(-z_n)}{e(z_n)}\right]\delta_{nm} + 2\varepsilon a - \lambda\rho\mathbb{D}(z_n, z_m) + \sum_{j=1}^{2}\gamma_j\, g_j(z_n)g_j(z_m). \tag{33}$$

If $s_1 = s_2$, the sum over j is replaced by its limit $\lambda\rho\,\partial_s[Z(s)/((z_n^2 - s)(z_m^2 - s))]$ at $s = s_1$, where $Z(s) \coloneqq [Z(u) + Z(-u)]/2$ with $u^2 = s$, and $Z(u)$ is Z_j with u_j replaced by u. For $r = \mathrm{m}$ and $k = 0$ the atomic pair is omitted, both from the sum and from the assumption $K_j \neq 0$. The formula does not depend on the sign chosen for u_j, and the inverse χ_r^{-1} it represents is two-sided.

So Γ_d and Γ_m have the same anatomy as Γ_s: a diagonal term, the constant $\pm U$, a dense term without thermal factors, and two separable terms with one thermal denominator K_j each. Only $\alpha_k(u_j)$, i.e. tanh or coth at the two roots, has to be evaluated. The assumption $e(z_n) \neq 0$ is not essential either. As e has real coefficients and $z_{\overline{n}} = z_n^*$, lattice zeros of e come in mirror orbits $\{n, \overline{n}\}$. If e vanishes on one orbit, Eq. 33 still holds, with the entries in the rows and columns of the orbit taken as limits as μ approaches its value (Appendix B).

At $\mu = 0$, $\rho = 0$ and Theorem 4 does not apply literally. Here the Cauchy matrix is the mirror, $C = h\mathcal{M}$ with $h = \overline{c} = \beta p_0$, $H(z) = z^2 + q_1$ is even and $e(z) = \left(z^2 + q_1\right)^2 - 16a^4 p_0^2 = (z^2 - \hat{s})(z^2 + 4\varepsilon a^2(p_0 + p_1) + q_0)$ with

$$\hat{s} \coloneqq 4\varepsilon a^2(p_0 - p_1) - q_0, \qquad \hat{g}(z) \coloneqq \frac{1}{z^2 - \hat{s}}, \qquad \hat{D} \coloneqq \beta - \frac{2\varepsilon a S_k(\hat{u})}{\hat{u}}, \tag{34}$$

where $\hat{u}$ is either square root of $\hat{s}$; $\hat{D}$ is even in $\hat{u}$. Explicitly, $\hat{D} = \beta[1 + \varepsilon a \tanh(\beta\hat{u}/2)/\hat{u}]$ for even k, with the value $\beta(1 + \varepsilon a\beta/2)$ at $\hat{u} = 0$; for odd k, coth replaces tanh, and $\hat{u} = 0$ means $e(0) = 0$ at the mirror fixed point $z_n = 0$ (see below).

Corollary 5 (half filling). Let $\mu = 0$, $e(z_n) \neq 0$ for all n, and $\hat{D} \neq 0$. Then

$$\Gamma_{r,nm} = \beta P(z_n)\left[\frac{1}{G_2(z_n)} - \frac{H(z_n)}{e(z_n)}\right]\delta_{nm} + 2\varepsilon a - \lambda h F(z_n)\delta_{m\overline{n}} - \frac{2\varepsilon a\beta P(\hat{u})}{\hat{D}}\hat{g}(z_n)\hat{g}(z_m), \tag{35}$$

and the inverse χ_r^{-1} it represents is two-sided.

This is Eq. 19 of Ref. [1] for $r = \mathrm{d}, \mathrm{m}$ (Appendix D). Compared with Eq. 33, the dense term has become a mirror term, and only one separable term with one thermal denominator remains. The corollary is the limit $\mu \to 0$ of Theorem 4 (proof in Appendix B). The assumption $e(z_n) \neq 0$ matters only for $\beta H(z_n) = -\lambda h$ with $n \neq \overline{n}$, a genuine divergence (Section 4.3). At all other zeros of $e(z_n)$, including both signs at a mirror fixed point (odd k, $z_n = 0$), Eq. 35 holds after taking limits (Appendix B).

4.3 Finite boxes and divergences

Proposition 6 (finite boxes). Under the assumptions of Theorem 2, Theorem 3, Theorem 4 or Corollary 5, the restrictions of χ_r to the boxes $|2n + 1 + k| \leq 2M + 1$ are invertible for all sufficiently large M, and their inverses converge entrywise to $\chi_r^{-1} = \Gamma_r/\beta^2 + \eta_r \boldsymbol{B}^{-1}$.

Divergences. An entry of Γ_r can diverge only where a denominator in its closed form vanishes. Removable are the zeros of $t_n + A_\mathrm{s}$ (Theorem 3), of $s_1 - s_2$, and the lattice zeros of e on one mirror orbit for $\mu \neq 0$ (Theorem 4); at $\mu = 0$, the zeros of $e(z_n)$ with $\beta H(z_n) = \lambda h$, and both kinds at a mirror fixed point (Appendix B). Away from intersections of these exceptional sets, the remaining zeros are genuine poles under the following nonvanishing conditions:

- $K_j = 0$ with $u_j P(u_j) \neq 0$ and $s_1 \neq s_2$ (d, m). Then $\gamma_j K_j = 2\lambda\rho\beta u_j P(u_j) H(-u_j)/(s_j - s_\ell) \neq 0$, as $\alpha_k(u_j) \neq 0$. At $u_j = 0$ the zero of K_j can cancel, as at $\hat{u} = 0$ in Corollary 5.
- $\hat{D} = 0$ with $P(\hat{u}) \neq 0$ (d, m at $\mu = 0$).
- $\beta H(z_n) = -\lambda h$ with $n \neq \overline{n}$ (d, m at $\mu = 0$). There $N(z, \cdot)/P$ is even, so by Eq. 61 $P(z)\chi_r(\boldsymbol{e}_n - \boldsymbol{e}_{\overline{n}}) = -(\beta H(z_n) + \lambda h)(\boldsymbol{e}_n - \boldsymbol{e}_{\overline{n}}) = 0$: the mirror-odd vector is a null vector. For $r = \mathrm{d}$, $k = 0$, $n = 0$ this is $\beta U = 2\pi/\sqrt{3}$.
- $K_\mathrm{s} = 0$ with $\mathcal{P}(-A_\mathrm{s}) \neq 0$ (s).
- $t_n + a\rho b + a^2 = 0$ with $n \neq \overline{n}$ (t); the numerator is $a^2(1 - \rho^2)\mathcal{P}(t_n) \neq 0$.

Example. *Closed form against finite boxes* ($r = \mathrm{d}$, $U = 2$, $\beta = 3$, $\mu = 0.3$, $k = 1$). The inverses of χ_d on the boxes of Proposition 6 ($2M + 1$ frequencies, as k is odd) converge like $1/M$; the extrapolation $2\Gamma^{(256)} - \Gamma^{(128)}$ removes the leading error.

(n, m)	$M = 128$	$M = 256$	extrapolated	Eq. 33
$(0, 0)$	$25.9568 + 33.8867i$	$25.9592 + 33.8867i$	$25.96155 + 33.88673i$	$25.96157 + 33.88673i$
$(0, \overline{0})$	-16.2893	-16.2870	-16.28460	-16.28459
$(2, -1)$	$2.2186 + 0.0612i$	$2.2210 + 0.0612i$	$2.22334 + 0.06123i$	$2.22336 + 0.06123i$

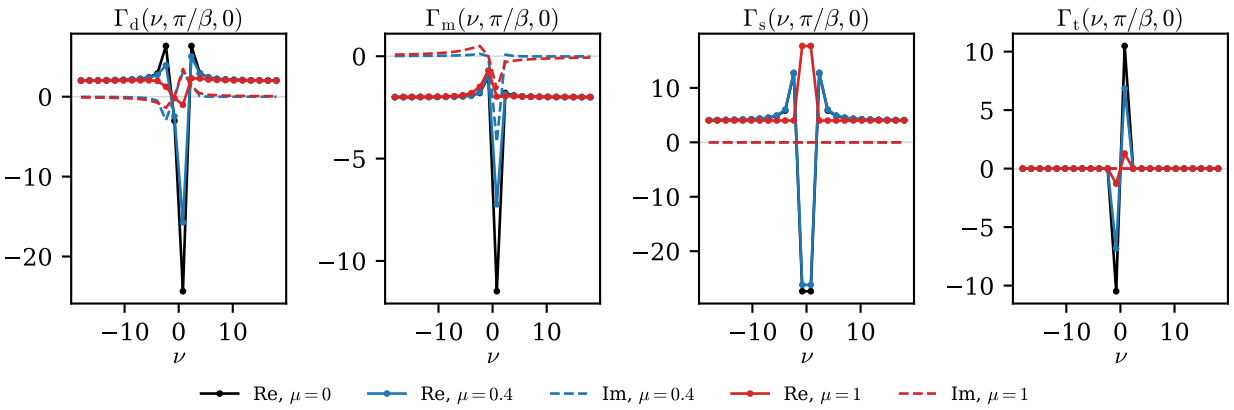


Figure 3: $\Gamma_r(\nu, \pi/\beta, 0)$ from Eq. 33 (Eq. 35 at $\mu = 0$), Eq. 27, Eq. 25 at $U = 2$, $\beta = 4$, for $\mu = 0$ (half filling, real), $\mu = 0.4$ and $\mu = 1$. Away from half filling Γ_d, Γ_m acquire imaginary parts; all Γ_r approach their bare values U, $-U$, $2U$, 0 at large $|\nu|$.

5 Matsubara sums

Lemma 7 (basic sum). For x off the lattice, $\mathrm{PV}\sum_n (z_n - x)^{-1} = S_k(x)$ (Eq. 22), and for $j \geq 2$, $\sum_n (z_n - x)^{-j} = S_k^{(j-1)}(x)/(j-1)!$, absolutely convergent. Moreover, for all k,

$$S_k(x \pm w/2) = -\frac{\beta}{2}\tanh(\beta(x-\mu)/2). \tag{36}$$

In particular, S_k takes the value $S_\pm := \pm(\beta/2)\tanh(\beta(a \pm \mu)/2)$ at both zeros $\mp a \pm w/2$ of $Q_\pm$; by Eq. 2, $S_+ = \beta(p_1 - p_0)/(2(p_1 + p_0))$ and $S_- = \beta(p_2 - p_1)/(2(p_1 + p_2))$.

Proof. For even k, $2n + 1 + k = 2n' + 1$ with $n' = n + k/2$, so $z_n - x = i\nu_{n'} - (x - \mu)$. For odd k, $2n + 1 + k = 2n''$, so $z_n - x = i\Omega_{n''} - (x - \mu)$. In both cases the box is symmetric in n' (resp. n''). The symmetric Mittag-Leffler expansions $\tanh y = \sum_j \big(y - i\pi\big(j + \tfrac{1}{2}\big)\big)^{-1}$ and $\coth y = \sum_j (y - i\pi j)^{-1}$ at $y = \beta(x - \mu)/2$ read $\tanh(\beta(x-\mu)/2) = (2/\beta)\sum_j \big(x - \mu - i\nu_j\big)^{-1}$ and $\coth(\beta(x-\mu)/2) = (2/\beta)\sum_j \big(x - \mu - i\Omega_j\big)^{-1}$, which is Eq. 22. For $j \geq 2$ the $(j-1)$-fold x-derivative of the series converges absolutely and uniformly on compact sets away from the lattice, so it may be differentiated term by term. For Eq. 36: $w/2 = i\pi k/\beta$ shifts the argument of tanh or coth by $\pm i\pi k/2$. For even k this is a multiple of the period $i\pi$; for odd k, $\coth(y \pm i\pi/2) = \tanh y$. □

Lemma 8 (residue rule). Let φ be proper rational without poles on the lattice. Then

$$\sum_n \varphi(z_n) = \sum_{x \in \text{ poles}(\varphi)} \operatorname*{Res}_{t=x}[\varphi(t)S_k(t)], \tag{37}$$

absolutely convergent if $\varphi = O(t^{-2})$, and as a PV sum if $\varphi \sim t^{-1}$. ($\mathrm{Res}_{t=x}$ of a function is the coefficient of $(t-x)^{-1}$ in its Laurent expansion at x.)

Proof. Partial fractions write φ as a finite sum of constants times $(t-x)^{-j}$, $j \geq 1$, over its poles x. By Lemma 7 each $(t-x)^{-j}$ sums to $S_k^{(j-1)}(x)/(j-1)!$, which is the residue of $(t-x)^{-j}S_k(t)$ at x, since S_k is analytic there. □

Lemma 9 (particle–particle sum). For $t \neq t_n$ $(\forall n)$, $\sum_n (t_n - t)^{-1} = \mathcal{T}_k(t)$ (Eq. 24), absolutely convergent. At the two points $t_\pm := -a^2 \pm ab$,

$$\mathcal{T}_k(t_+) = -\frac{2S_-}{2a - b}, \qquad \mathcal{T}_k(t_-) = \frac{2S_+}{2a + b}. \tag{38}$$

Proof. Since $x_n = z_n - w/2$ and $b/2 = \mu - w/2$, $t_n = b^2/4 - (z_n - \mu)^2$, so $(t_n - t)^{-1} = \big[(z_n - \mu + \xi)^{-1} - (z_n - \mu - \xi)^{-1}\big]/(2\xi)$. By Lemma 7 and the oddness of tanh and coth, the sum is $(S_k(\mu - \xi) - S_k(\mu + \xi))/(2\xi) = -S_k(\mu + \xi)/\xi$; it converges absolutely, as $t_n \sim n^2$. At $t_\pm$, $\xi^2 = (a \mp b/2)^2$. At t_+ take $\xi = a - b/2$: then $\mu + \xi = a + w/2$, and $S_k(\mu + \xi) = S_-$ by Eq. 36. At t_- take $\xi = -a - b/2$: then $\mu + \xi = -a + w/2$, and $S_k(\mu + \xi) = S_+$. □

6 Proofs for the singlet and triplet channels

6.1 Variables

With $t_\pm$ from Lemma 9, $\mathcal{P}(t) = (t - t_+)(t - t_-)$. Define

$$d_\pm := A_\mathrm{s} + t_\pm, \qquad c := c_{-k} = \begin{cases} \rho/b & (\mu, k) \neq (0, 0) \\ \beta p_0 & \mu = k = 0 \end{cases}. \tag{39}$$

Then $d_+ d_- = \mathcal{P}(-A_\mathrm{s})$. For $t = x(b - x)$, direct expansion gives

$$(x-a)(b-x-a)=t-t_+,\qquad (x+a)(b-x+a)=t-t_-,$$
$$(x+a\rho)(b-x+a\rho)=\mathcal{G}(t), \tag{40}$$

and the product of the first two is $(x^2-a^2)((b-x)^2-a^2)=\mathcal{P}(t)$. Hence, by Eq. 10,

$$B_n=-\frac{\beta}{2}\frac{\mathcal{G}(t_n)}{\mathcal{P}(t_n)}=B_{\overline{n}}. \tag{41}$$

The pp entry (n,m) at transfer k is the ph entry at bosonic index $k':=-k-n-m-1$, with frequency $w':=i\omega_{k'}=b-x_n-x_m$. Then $\delta_{k'0}=\delta_{m\overline{n}}$, and the pair coefficient becomes $c_{n+m+k'+1}=c$. In the variables of Lemma 1 at transfer w', $z+v=x_n+x_m+w'=b$. By Eq. 18 and Eq. 40,

$$Q_-(z)Q_-(v)=(x_n-a)(b-x_n-a)(x_m-a)(b-x_m-a)=(t_n-t_+)(t_m-t_+), \tag{42}$$

and likewise $Q_+(z)Q_+(v)=(t_n-t_-)(t_m-t_-)$.

6.2 Structure

Let $\mathcal{M}_\pm:=(\mathbb{1}\pm\mathcal{M})/2$ be the projectors onto mirror-even and mirror-odd vectors. A diagonal matrix whose entries depend on n only through t_n commutes with the mirror $\mathcal{M}$. Define

$$L_{\mathrm{t}}(t):=-\frac{\beta}{2}\frac{t+a\rho b+a^2}{\mathcal{P}(t)},\quad L_{\mathrm{s}}(t):=\frac{\beta}{2}\frac{t+A_{\mathrm{s}}}{\mathcal{P}(t)},\quad g_\pm(t):=\frac{1}{t-t_\pm},\quad \gamma_\pm:=\frac{a(\rho\pm1)(2a\mp b)}{2b}, \tag{43}$$

and $\boldsymbol{L}_{\mathrm{s,\,t}}:=\mathrm{diag}\big(L_{\mathrm{s,\,t}}(t_n)\big)$.

Proposition 10.

$$\chi_{\mathrm{t}}=\boldsymbol{B}\mathcal{M}_++\boldsymbol{L}_{\mathrm{t}}\mathcal{M}_-,\qquad \chi_{\mathrm{s}}=\boldsymbol{L}_{\mathrm{s}}\mathcal{M}_+-\boldsymbol{B}\mathcal{M}_-+\chi_{\mathrm{s,\,reg}}, \tag{44}$$

where

$$\big(\chi_{\mathrm{s,\,reg}}\big)_{nm}=\frac{a(2ac-1)(t_n-t_-)(t_m-t_-)+a^2(1-\rho)(b+2a)(t_n+t_m+2a^2)}{\mathcal{P}(t_n)\mathcal{P}(t_m)}, \tag{45}$$

which for $b\neq0$ equals

$$\big(\chi_{\mathrm{s,\,reg}}\big)_{nm}=\gamma_+g_+(t_n)g_+(t_m)+\gamma_-g_-(t_n)g_-(t_m). \tag{46}$$

Proof. Insert $w\to w'$ into Eq. 13 and Eq. 14. In the δ_{nm} terms, $x_n+w'=b-x_m=x_{\overline{n}}$; in the $\delta_{k'0}=\delta_{m\overline{n}}$ terms, $x_m=x_{\overline{n}}$ and $x_n+w'=x_n$. With Eq. 40 (abbreviating $\mathcal{P}:=\mathcal{P}(t_n)$),

$$\begin{aligned}\chi_{\mathrm{pp},\uparrow\uparrow}&=2B_n\delta_{nm}-\frac{\beta a^2(1-\rho^2)}{\mathcal{P}}(\delta_{nm}-\delta_{m\overline{n}}),\\ \chi_{\mathrm{pp},\uparrow\downarrow}&=-\frac{4\beta a^2p_1}{\mathcal{P}}\delta_{nm}+\frac{4\beta a^2(p_0p_2-p_1^2)}{\mathcal{P}}\delta_{m\overline{n}}+\mathcal{N}|_{w=w'}.\end{aligned} \tag{47}$$

Triplet. $\big(\chi_{\mathrm{pp},\uparrow\uparrow}+2\boldsymbol{B}\big)/4$ has δ_{nm} coefficient $B_n-\beta a^2(1-\rho^2)/(4\mathcal{P})$ and $\delta_{m\overline{n}}$ coefficient $\beta a^2(1-\rho^2)/(4\mathcal{P})$. Using Eq. 41, these equal $(B+L_{\mathrm{t}})/2$ and $(B-L_{\mathrm{t}})/2$. As B and L_{t} commute with $\mathcal{M}$, $\frac{1}{2}(B+L_{\mathrm{t}})+\frac{1}{2}(B-L_{\mathrm{t}})\mathcal{M}=B\mathcal{M}_++L_{\mathrm{t}}\mathcal{M}_-$.

Singlet, delta terms. The δ_{nm} coefficient of $\big(-\chi_{\mathrm{pp},\uparrow\uparrow}+2\chi_{\mathrm{pp},\uparrow\downarrow}-2\boldsymbol{B}\big)/4$ is $-B_n+\beta a^2(1-\rho^2-8p_1)/(4\mathcal{P})=(L_{\mathrm{s}}-B)/2$. The $\delta_{m\overline{n}}$ coefficient is $\beta a^2[-(1-\rho^2)+8(p_0p_2-p_1^2)]/(4\mathcal{P})$. By Eq. 4 and $p_0+p_2=1-2p_1$, the bracket is $1-8p_1-\rho^2$, and the coefficient is $(L_{\mathrm{s}}+B)/2$. Together: $L_{\mathrm{s}}\mathcal{M}_+-B\mathcal{M}_-$.

Singlet, remaining terms: $(2/4)\mathcal{N}|_{w=w'}$. Insert into Eq. 19: $4a^2C\to4a^2c$, $z+v=b$, $zv+q_0=a^2+(t_n+t_m)/2$ at transfer w' (identity (A3)), and Eq. 42. This gives Eq. 45; for $b\neq0$, its partial fractions in t_n and t_m are Eq. 46 (identity (A3)). □

6.3 Proofs

Proof of Theorem 2. χ_{t} and $\boldsymbol{B}$ act on each orbit $\{n,\overline{n}\}$ separately. For $n \neq \overline{n}$ the orbit carries a 2×2 block with eigenvalue B_n on the even vector and $L_{\mathrm{t}}(t_n)$ on the odd vector; $L_{\mathrm{t}}(t_n) \neq 0$ is the assumption $t_n + a\rho b + a^2 \neq 0$. So $\chi_{\mathrm{t}}^{-1} = \boldsymbol{B}^{-1}\mathcal{M}_+ + \boldsymbol{L}_{\mathrm{t}}^{-1}\mathcal{M}_-$, and $\Gamma_{\mathrm{t}} = \beta^2(\boldsymbol{L}_{\mathrm{t}}^{-1} - \boldsymbol{B}^{-1})\mathcal{M}_-$ by Eq. 11. Further, $(\beta^2/2)(1/L_{\mathrm{t}} - 1/B) = -\beta\mathcal{P}[\mathcal{G} - (t + a\rho b + a^2)]/((t + a\rho b + a^2)\mathcal{G})$ with $\mathcal{G} - (t + a\rho b + a^2) = -a^2(1-\rho^2)$. At a fixed point the orbit is $\{n\}$ with $\mathcal{M}_- = 0$; then $\chi_{\mathrm{t},nn} = B_n$, $\Gamma_{\mathrm{t},nn} = 0$, and $L_{\mathrm{t}}(t_n)$ does not enter. □

Proof of Theorem 3. Idea. χ_{s} is an easily invertible matrix plus the rank-two term Eq. 46, so the Woodbury formula applies. Its 2×2 matrix becomes diagonal in a suitable basis.

For $U = 0$ both sides vanish ($L_{\mathrm{s}} = -B$, $\chi_{\mathrm{s,\,reg}} = 0$). Let $U \neq 0$, and first assume $b \neq 0$, $b \neq \pm 2a$, $d_\pm \neq 0$ and $t_n + A_{\mathrm{s}} \neq 0$ for all n. Then $\gamma_\pm \neq 0$, since $|\rho| < 1$.

Woodbury. Let $\boldsymbol{g}$ be the $\infty \times 2$ matrix with rows $(g_+(t_n), g_-(t_n))$ and $\gamma := \mathrm{diag}(\gamma_+, \gamma_-)$, so $\chi_{\mathrm{s,\,reg}} = \boldsymbol{g}\gamma\boldsymbol{g}^\top$ and $\mathcal{M}_+\boldsymbol{g} = \boldsymbol{g}$. The matrix $\boldsymbol{X} := \boldsymbol{L}_{\mathrm{s}}\mathcal{M}_+ - \boldsymbol{B}\mathcal{M}_-$ has inverse $\boldsymbol{X}^{-1} = \boldsymbol{L}_{\mathrm{s}}^{-1}\mathcal{M}_+ - \boldsymbol{B}^{-1}\mathcal{M}_-$ (all products are finite sums). Set $\boldsymbol{V}_{\mathrm{s}} := \boldsymbol{X}^{-1}\boldsymbol{g} = \boldsymbol{L}_{\mathrm{s}}^{-1}\boldsymbol{g}$ and $\Psi := \boldsymbol{g}^\top\boldsymbol{V}_{\mathrm{s}} = \sum_n (g_+, g_-)^\top(g_+, g_-)(t_n)/L_{\mathrm{s}}(t_n)$, absolutely convergent (terms $O(t_n^{-1}) = O(n^{-2})$). If $\gamma^{-1} + \Psi$ is invertible, then

$$Y := \boldsymbol{X}^{-1} - \boldsymbol{V}_{\mathrm{s}}(\gamma^{-1} + \Psi)^{-1}\boldsymbol{V}_{\mathrm{s}}^\top \tag{48}$$

satisfies $\chi_{\mathrm{s}}Y = \mathbb{1} + \boldsymbol{g}\Big[\gamma - (\mathbb{1} + \gamma\Psi)(\gamma^{-1} + \Psi)^{-1}\Big]\boldsymbol{V}_{\mathrm{s}}^\top = \mathbb{1}$. Y is symmetric, so also $Y\chi_{\mathrm{s}} = \mathbb{1}$. By Eq. 11 ($\eta_{\mathrm{s}} = -1$),

$$\Gamma_{\mathrm{s}} = \beta^2(\boldsymbol{L}_{\mathrm{s}}^{-1} + \boldsymbol{B}^{-1})\mathcal{M}_+ - \beta^2\boldsymbol{V}_{\mathrm{s}}(\gamma^{-1} + \Psi)^{-1}\boldsymbol{V}_{\mathrm{s}}^\top. \tag{49}$$

The first term gives the first term of Eq. 27: $(\beta^2/2)(1/L_{\mathrm{s}} + 1/B) = \beta\mathcal{P}[1/(t + A_{\mathrm{s}}) - 1/\mathcal{G}]$ and $\mathcal{G} - t - A_{\mathrm{s}} = a^2(\rho^2 - 1 + 8p_1)$.

Change of basis. Let $\kappa_n := 1/(t_n + A_{\mathrm{s}})$. Since $\mathcal{P} = (t - t_+)(t - t_-)$ and $t - t_\mp = (t + A_{\mathrm{s}}) - d_\mp$,

$$\frac{g_\pm(t_n)}{L_{\mathrm{s}}(t_n)} = \frac{2}{\beta}(1 - d_\mp\kappa_n), \quad \text{i.e.} \quad (\boldsymbol{V}_{\mathrm{s}})^\top_{n,:} = \frac{2}{\beta}\,\Theta^\top(1, \kappa_n)^\top, \quad \Theta := \begin{pmatrix} 1 & 1 \\ -d_- & -d_+ \end{pmatrix}. \tag{50}$$

Hence $\Big(\boldsymbol{V}_{\mathrm{s}}(\gamma^{-1} + \Psi)^{-1}\boldsymbol{V}_{\mathrm{s}}^\top\Big)_{nm} = (2/\beta)^2(1, \kappa_n)\Xi^{-1}(1, \kappa_m)^\top$ with $\Xi := \Theta^{-\top}(\gamma^{-1} + \Psi)\Theta^{-1}$. Inserting $\Psi = \sum_n (2/\beta)^2 L_{\mathrm{s}}(t_n)\Theta^\top(1, \kappa_n)^\top(1, \kappa_n)\Theta$ and $(2/\beta)^2 L_{\mathrm{s}} = (2/\beta)(t + A_{\mathrm{s}})/\mathcal{P}$,

$$\Xi = \Theta^{-\top}\gamma^{-1}\Theta^{-1} + \frac{2}{\beta}\sum_n \frac{1}{\mathcal{P}(t_n)}\begin{pmatrix} t_n + A_{\mathrm{s}} & 1 \\ 1 & 1/(t_n + A_{\mathrm{s}}) \end{pmatrix}. \tag{51}$$

By partial fractions over $t_+, t_-, -A_{\mathrm{s}}$, the three sums are combinations of $\mathcal{T}_k(t_+)$, $\mathcal{T}_k(t_-)$ and $\mathcal{T}_k(-A_{\mathrm{s}})$. Inserting Eq. 38, the off-diagonal entries of the two terms cancel, and (identity (A4))

$$\Xi = \mathrm{diag}\left(-\frac{1}{a}, \frac{K_{\mathrm{s}}}{a d_+ d_-}\right). \tag{52}$$

So $\gamma^{-1} + \Psi$ is invertible iff $K_{\mathrm{s}} \neq 0$, and

$$-\beta^2\left(\frac{2}{\beta}\right)^2(1, \kappa_n)\Xi^{-1}(1, \kappa_m)^\top = -4\left[-a + \frac{a d_+ d_- \kappa_n\kappa_m}{K_{\mathrm{s}}}\right] = 4a - \frac{4a\,\mathcal{P}(-A_{\mathrm{s}})}{K_{\mathrm{s}}(t_n + A_{\mathrm{s}})(t_m + A_{\mathrm{s}})}. \tag{53}$$

Excluded cases. Fix k. The values of μ where $b = 0$, $b = \pm 2a$, $d_\pm = 0$ or $t_n + A_{\mathrm{s}} = 0$ for some n are isolated, since these are nonconstant analytic functions of μ. Eq. 27 contains no b, $2a \pm b$ or $d_\pm$ in a denominator, and its divergent parts at $q := t_n + A_{\mathrm{s}} \to 0$ cancel. Indeed, the orbit $\{n, \overline{n}\}$ has $r \in \{1, 2\}$ elements, so $\mathcal{T}_k(-A_{\mathrm{s}}) = r/q + O(1)$ and $K_{\mathrm{s}} = 2ar/(\beta q) + O(1)$. Since $a^2(\rho^2 - 1 + 8p_1) = \mathcal{G}(t_n) - q$ and $\mathcal{G}(t_n) \neq 0$ (Eq. 40), the first term of Eq. 27 is $\beta\mathcal{P}(-A_{\mathrm{s}})(\delta_{nm} + \delta_{m\overline{n}})/q + O(1)$, and

the last one is $-2\beta\mathcal{P}(-A_{\mathrm{s}})/(rq)+O(1)$ for $m\in\{n,\overline{n}\}$ and $O(1)$ otherwise. For $m\in\{n,\overline{n}\}$, $\delta_{nm}+\delta_{m\overline{n}}=2/r$, so the $1/q$ terms cancel. Near all these values, $(\chi_{\mathrm{s}})_{nl}=O(|t_l|^{-1})$ and Y_{lm} is bounded for $l\notin\{n,\overline{n},m,\overline{m}\}$, uniformly, where $Y=\Gamma_{\mathrm{s}}/\beta^2-\boldsymbol{B}^{-1}$ with Eq. 27. As $|t_l|\sim l^2$, Lemma 17 extends $\chi_{\mathrm{s}}Y=\mathbb{1}$ to these values. □

7 The Cauchy matrix

Throughout this section $\mu\neq 0$.

Lemma 11 (values on the reflected lattice). For every n,

$$S_k(-z_n)=\frac{\beta}{2}\coth(\beta\mu),\qquad S_k'(-z_n)=\frac{\beta^2}{4\sinh^2(\beta\mu)},\qquad \rho\,S_k(-z_n)=\overline{c}. \tag{54}$$

Proof. Write $-z_n=x-w/2$ with $x:=-\mu-i\pi(2n+1)/\beta$. By Eq. 36, $S_k(-z_n)=-(\beta/2)\tanh(-\beta\mu-i\pi(2n+1)/2)=(\beta/2)\coth(\beta\mu)$, as tanh has period $i\pi$ and $\tanh(y-i\pi/2)=\coth y$. Differentiating Eq. 36 in x gives $S_k'(-z_n)=-(\beta^2/4)\cosh^{-2}(-\beta\mu-i\pi(2n+1)/2)=\beta^2/(4\sinh^2(\beta\mu))$. The last identity is Eq. 5. □

So S_k is *constant* on the reflected lattice (Figure 2). This is the source of all simplifications below.

Lemma 12 (square). $\sum_l C_{nl}C_{lm}=h^2\delta_{nm}$, absolutely convergent.

Proof. $C_{nl}C_{lm}=O(|z_l|^{-2})$. For $n\neq m$, partial fractions $[(z_n+z_l)(z_l+z_m)]^{-1}=(z_m-z_n)^{-1}\left[(z_l+z_n)^{-1}-(z_l+z_m)^{-1}\right]$ and Lemma 7 give $\rho^2[S_k(-z_n)-S_k(-z_m)]/(z_m-z_n)=0$ by Lemma 11. For $n=m$, $\rho^2\sum_l(z_l+z_n)^{-2}=\rho^2S_k'(-z_n)=h^2$ by Lemma 11 and Eq. 5. □

The action of C on rational functions is governed by the thermal function α_k of Eq. 30.

Lemma 13 (thermal function). With $y_\pm:=\beta(x\pm\mu)/2$,

$$\alpha_k(x)=\begin{cases}-h\cosh(y_+)/\cosh(y_-) & k \text{ even}\\ -h\sinh(y_+)/\sinh(y_-) & k \text{ odd}\end{cases},\qquad \alpha_k(x)\alpha_k(-x)=h^2. \tag{55}$$

Proof. For even k, insert $S_k(x)=-(\beta/2)\tanh y_-$, $p_2=p_0e^{2\beta\mu}$ and $h=\beta p_0e^{\beta\mu}$:

$$\alpha_k(x)=-\frac{\beta}{2}\frac{p_2e^{y_-}+p_0e^{-y_-}}{\cosh y_-}=-h\frac{\cosh(y_-+\beta\mu)}{\cosh y_-},\quad y_-+\beta\mu=y_+. \tag{56}$$

Under $x\to-x$, $y_\pm\to-y_\mp$, so the two quotients multiply to 1. Odd k: replace tanh, cosh by coth, sinh. □

Let $\mathcal{C}$ be the action of C on functions sampled on the lattice, $(\mathcal{C}\,\varphi)(z_m):=\sum_n C_{mn}\varphi(z_n)$.

Lemma 14 (Cauchy calculus). For x with x and $-x$ off the lattice, and every lattice point z,

$$\mathcal{C}[(t-x)^{-1}](z)=\frac{\alpha_k(x)}{z+x},\qquad \mathcal{C}[1](z)=\overline{c}\quad(\mathrm{PV}). \tag{57}$$

Higher poles follow by applying $(1/j!)\partial_x^j$; so $\mathcal{C}$ maps every proper rational function whose poles x satisfy this condition to an explicit proper rational function.

Proof. Partial fractions and Lemma 7 give

$$\sum_n\frac{\rho}{(z+z_n)(z_n-x)}=\frac{\rho[S_k(x)-S_k(-z)]}{z+x}=\frac{\alpha_k(x)}{z+x} \tag{58}$$

by Lemma 11. The constant rule is $\mathrm{PV}\sum_n C_{mn} = \rho S_k(-z_m) = \overline{c}$. The sums of x-derivatives converge uniformly near any admissible x, so Eq. 57 may be differentiated. □

8 Proofs for the density and magnetic channels

8.1 Structure

Let $\delta := \delta_{k0}$ and

$$\zeta_r := \beta a^2[1-\rho^2+4\varepsilon(p_0p_2-p_1^2)], \qquad N_2(v) := -2\varepsilon a(v^2+2a\rho v+q_0)+\delta\zeta_r(v^2-a^2), \tag{59}$$

$$N_0(v) := (v^2+2q_0-4a^2)N_2(v)+(2\varepsilon a-\delta\zeta_r)P(v), \qquad N(z,v) := N_0(v)+z^2N_2(v). \tag{60}$$

P is even by Eq. 18 and has no zeros on the lattice. In N_0 the terms of degree four cancel, so $N_0(v)/P(v) = O(v^{-1})$ and $N_2(v)/P(v) = O(v^{-2})$. We identify a function φ on the lattice with the vector $(\varphi(z_n))_n$ and write $(\chi_r\varphi)(z_n) := \sum_m \chi_{r,nm}\varphi(z_m)$.

Proposition 15 (structure).

$$P(z_n)\chi_{r,nm} = -\beta H(z_n)\delta_{nm} + \lambda C_{nm} + \frac{N(z_n,z_m)}{P(z_m)}. \tag{61}$$

Proof. Write $z := z_n$, $v := z_m$, and collect the terms of $\chi_r = \chi_{\uparrow\uparrow} + \varepsilon\chi_{\uparrow\downarrow}$ (Eq. 13, Eq. 14).

(a) δ_{nm} terms: $-\beta[G(x_n)G(x_n+w)+a^2(1-\rho^2+4\varepsilon p_1)J(x_n)J(x_n+w)]$. By Eq. 7 and Eq. 18, $G(x_n)G(x_n+w) = [(z+a\rho)^2-w^2/4]/P(z)$ and $J(x_n)J(x_n+w) = 1/P(z)$. The numerators add up to $(z+a\rho)^2 - w^2/4 + a^2(1-\rho^2) + 4\varepsilon a^2 p_1 = H(z)$.

(b) δ_{k0} terms (only for $k=0$, where $w=0$, $x_n=z$ and $P(z) = (z^2-a^2)^2$): $\zeta_r J(z)J(v)$, and $P(z)J(z)J(v) = (z^2-a^2)(v^2-a^2)/P(v)$.

(c) Remaining terms: $\varepsilon\mathcal{N}_{nm}$ with Eq. 19. Since $C_{nm}(z+v) = \rho$ and $Q_+(z)-Q_-(v) = (z+v)(z-v+2a)$, we have $Q_+(z)C_{nm} = Q_-(v)C_{nm} + \rho(z-v+2a)$. With $1/Q_-(v) = Q_+(v)/P(v)$ this gives

$$\begin{aligned} P(z)\mathcal{N}_{nm} = 4a^2C_{nm} + \frac{1}{P(v)}[&4a^2\rho(z-v+2a)Q_+(v) - 2aQ_+(z)Q_+(v) \\ &+4a^2(1-\rho)(z+v+2a)(zv+q_0)]. \end{aligned} \tag{62}$$

The numerator is quadratic in z. Its z coefficient is $4a^2Q_+(v)[\rho-1+(1-\rho)] = 0$, because $v^2+2av+q_0 = Q_+(v)$. Its z^2 coefficient is $-2aQ_+(v)+4a^2(1-\rho)v = -2a(v^2+2a\rho v+q_0)$, and its z^0 coefficient is $-2a(v^2+2q_0-4a^2)(v^2+2a\rho v+q_0)+2aP(v)$ (identity (A2)). Times ε, this is the part of Eq. 60 without ζ_r. The term (b) is the rest, since $(v^2-2a^2)(v^2-a^2)-P(v) = -a^2(v^2-a^2)$ at $k=0$. □

So, after multiplication by P, the finite-rank part of χ_r has only two output directions, 1 and z^2.

8.2 Rank-two sums

Lemma 16 (rank-two sums). For x off the lattice with $P(x) \neq 0$, and every z,

$$\mathcal{I}(z;x) := \sum_l \frac{N(z,z_l)}{P(z_l)(z_l-x)} = \overline{\mathcal{I}}(z;x) + \alpha_k(x)\frac{N(z,x)}{\rho P(x)}, \tag{63}$$

absolutely convergent, where

$$\overline{\mathcal{I}}(z;x) := \frac{\overline{c}N(z,x)}{\rho P(x)} + \sum_{y:P(y)=0} \operatorname*{Res}_{t=y} \frac{S_k(t)N(z,t)}{P(t)(t-x)}. \tag{64}$$

$\overline{\mathcal{I}}$ depends on the temperature only through β, $\overline{c}$ and $S_\pm$ (Lemma 7), and $P(x)\overline{\mathcal{I}}(z;x)$ is a polynomial in z and x.

Proof. At the zeros of P, $S_k = S_\pm$ (Lemma 7); for $k = 0$ they are double, and there $S_0' = S_0^2 - \beta^2/4$ by Eq. 36. The summand of $\mathcal{I}$ is $O(z_l^{-2})$, since N_0 has degree at most three. Lemma 8 gives the residue $S_k(x)N(z,x)/P(x)$ at $t = x$, plus the residues at the zeros of P; insert $\rho S_k(x) = \overline{c} + \alpha_k(x)$ (Eq. 30). In x, $\overline{\mathcal{I}}$ has poles only at the zeros of P, of at most their multiplicity. □

8.3 The proof of Theorem 4

Strategy. For $U = 0$, $\chi_r = \boldsymbol{B}$ and $\Gamma_\mathrm{d} = \Gamma_\mathrm{m} = 0$; let $U \neq 0$. Fix a lattice point $v = z_m$. We construct a bounded function y_v on the lattice with $\chi_r y_v = \boldsymbol{e}_m$, the m-th unit vector. By Eq. 61 this means, for every lattice point z,

$$-\beta H(z) y_v(z) + \lambda(\mathcal{C}\, y_v)(z) + \sum_l \frac{N(z,z_l)}{P(z_l)} y_v(z_l) = P(v)\delta_{zv}. \tag{65}$$

For $l \neq n$, $\chi_{r,nl} = O(|z_l|^{-2})$ by Eq. 13, Eq. 14 and Eq. 19, so $\sum_l \chi_{r,nl} y_v(z_l)$ converges absolutely. Its parts in Eq. 65 converge at least as PV sums (Lemma 14, Lemma 8), so they may be evaluated separately. We use Lemma 16 at $x = -v$, and Lemma 14 and Lemma 16 at $x = \pm u_j$. These points are off the lattice ($\mu \neq 0$, $e(z_n) \neq 0$), and $P(-v) = P(v) \neq 0$. We first assume that $u_j \neq 0$ and $P(u_j) \neq 0$ for the non-atomic pairs; Step 5 removes this assumption.

Step 1 (the Cauchy column). Through C, a unit vector at v produces the column $\lambda\rho/(z+v)$. Conversely, C maps $1/(t+v)$ back to a unit vector: $\mathcal{C}[(t+v)^{-1}](z) = \rho^{-1}\sum_l C(z,z_l)C(z_l,v) = (h^2/\rho)\delta_{zv}$ by Lemma 12. So try $y = D_v\delta_{zv} + E_v/(z+v)$. Insert it into the left side of Eq. 65. Use $H(z) = H(-v) + (z+v)(z-v+2a\rho)$, and Eq. 63 at $x = -v$, where $\alpha_k(-v) = 0$ (Lemma 11). The left side becomes

$$\begin{aligned}&\left[-\beta H(v) D_v + \frac{\lambda h^2}{\rho} E_v\right]\delta_{zv} + \frac{\lambda\rho D_v - \beta H(-v) E_v}{z+v} \\ &-\beta E_v(z-v+2a\rho) + D_v\frac{N(z,v)}{P(v)} + E_v\overline{\mathcal{I}}(z;-v).\end{aligned} \tag{66}$$

We choose D_v, E_v such that the first term is $P(v)\delta_{zv}$ and the second vanishes. The determinant of these two linear equations is $\beta^2 H(v)H(-v) - \lambda^2 h^2 = \beta^2 e(v) \neq 0$, and the solution is

$$D_v = -\frac{H(-v)P(v)}{\beta e(v)}, \qquad E_v = -\frac{\lambda\rho F(v)}{\beta^2}. \tag{67}$$

With these values the remaining terms are $1/(\beta e(v))$ times the left side of the identity (A7),

$$\begin{aligned}&\lambda\rho(z-v+2a\rho)P(v) - H(-v)N(z,v) - \frac{\lambda\rho}{\beta}P(v)\overline{\mathcal{I}}(z;-v) \\ &\quad = -e(v)N_2(z) + \lambda\rho z[P(v) - e(v)] + \lambda W(z;v^2).\end{aligned} \tag{68}$$

Here $W(z;s) \coloneqq W_0(s) + z^2 W_2(s)$ with polynomials W_0, W_2 of degree one in s, given in Appendix A. Both sides of Eq. 68 are polynomials in z and v (Lemma 16). So the remaining terms are the polynomial

$$\frac{1}{\beta}\left[-N_2(z) + \lambda\rho z(F(v) - 1) + \lambda\frac{W(z;v^2)}{e(v)}\right]. \tag{69}$$

Step 2 (the constant). For $y = 1$ the left side of Eq. 65 is $-\beta H(z) + \lambda\overline{c} + \mathrm{PV}\sum_l N(z,z_l)/P(z_l) = \beta N_2(z)/(2\varepsilon a)$, by $\mathcal{C}[1] = \overline{c}$ (Lemma 14) and identity (A6). So adding the constant $2\varepsilon a/\beta^2$ to y removes the first term of Eq. 69.

Step 3 (pole pairs). The rest of Eq. 69 must be removed by functions that are mapped to polynomials. A simple pole $1/(t-x)$ is mapped by $-\beta H + \lambda\,\mathcal{C}$ to poles at x and at $-x$ (Lemma 14). So consider $\varphi = f_+/(t-x) + f_-/(t+x)$. For it, the left side of Eq. 65 is a polynomial plus

$$\frac{-\beta H(x) f_+ + \lambda \alpha_k(-x) f_-}{z-x} + \frac{\lambda \alpha_k(x) f_+ - \beta H(-x) f_-}{z+x}. \tag{70}$$

Both residues vanish for some $(f_+, f_-) \neq 0$ iff the determinant $\beta^2 H(x)H(-x) - \lambda^2 \alpha_k(x)\alpha_k(-x) = \beta^2 e(x)$ vanishes (Eq. 55), i.e. iff $x = \pm u_j$. For $x = u_j$ we take

$$\varphi_j(t) := \frac{\beta H(-u_j)}{t-u_j} + \frac{\lambda \alpha_j}{t+u_j} = \frac{K_j t + u_j\big[\beta H(-u_j) - \lambda \alpha_j\big]}{t^2 - s_j}, \qquad \alpha_j := \alpha_k(u_j). \tag{71}$$

With $H(z) - H(\pm u_j) = (z \mp u_j)(z \pm u_j + 2a\rho)$, the polynomial part is

$$-\beta\big[K_j(z+2a\rho) + u_j(\beta H(-u_j) - \lambda\alpha_j)\big] + \beta H(-u_j)\mathcal{I}(z;u_j) + \lambda\alpha_j \mathcal{I}(z;-u_j). \tag{72}$$

To evaluate it, set $v = \mp u_j$ in Eq. 68, where $e(v) = 0$ and $P(v) = P(u_j)$:

$$\frac{\lambda\rho}{\beta} P(u_j)\overline{\mathcal{I}}(z;\pm u_j) = \lambda\rho(\pm u_j + 2a\rho)P(u_j) - H(\pm u_j)N(z, \mp u_j) - \lambda W(z;s_j). \tag{73}$$

Insert this into Eq. 63 for $\mathcal{I}(z;\pm u_j)$, with $\alpha_k(-u_j) = h^2/\alpha_j$ (Eq. 55):

- The terms with $N(z,-u_j)$ carry the factor $\lambda^2 h^2 - \beta^2 H(u_j)H(-u_j) = -\beta^2 e(u_j) = 0$.
- The terms with $N(z,u_j)$ cancel.
- The terms with $\lambda\rho P(u_j)$ cancel the constant part of the first bracket.

What remains is

$$P(z)(\chi_r \varphi_j)(z) = -\beta K_j \left[z + \frac{W(z;s_j)}{\rho P(u_j)} \right]. \tag{74}$$

Step 4 (solving). P and e are monic and even of degree four, and $W(z;s)$ has degree one in s. Partial fractions in v^2 therefore give

$$F(v) - 1 = \sum_j F_j g_j(v), \qquad \frac{W(z;v^2)}{e(v)} = \sum_j \frac{W(z;s_j)}{s_j - s_\ell} g_j(v), \qquad F_j := \frac{P(u_j)}{s_j - s_\ell}. \tag{75}$$

So the last two terms of Eq. 69 are $\beta^{-1}\sum_j \lambda\rho F_j g_j(v)\big[z + W(z;s_j)/\big(\rho P(u_j)\big)\big]$. By Steps 1–3 and Eq. 74,

$$y_v(z) := D_v \delta_{zv} + \frac{E_v}{z+v} + \frac{1}{\beta^2}\left[2\varepsilon a + \lambda\rho \sum_j \frac{F_j g_j(v)}{K_j} \varphi_j(z) \right] \tag{76}$$

solves Eq. 65. To read off Γ_r, note that $\lambda\rho F_j g_j(v)\varphi_j(z)/K_j = \lambda\rho F_j z g_j(z) g_j(v) + \gamma_j g_j(z) g_j(v)$ by Eq. 71. Further, $F(z) - F(v) = -(z^2 - v^2)\sum_j F_j g_j(z) g_j(v)$ gives

$$\mathbb{D}(z,v) = \frac{F(v)}{z+v} - z\sum_j F_j g_j(z) g_j(v). \tag{77}$$

Hence

$$y_v(z) = -\frac{H(-z)P(z)}{\beta e(z)}\delta_{zv} + \frac{1}{\beta^2}\left[2\varepsilon a - \lambda\rho\mathbb{D}(z,v) + \sum_j \gamma_j g_j(z) g_j(v) \right]. \tag{78}$$

By Eq. 7 and Eq. 18, $B_n = -\beta G_2(z_n)/P(z_n)$; so by Eq. 11, $\beta^2\big[y_{z_m}(z_n) - \delta_{nm}/B_n\big]$ is the right side of Eq. 33.

Step 5 (two-sidedness, branches, special cases).

- Let $Y_{nm} \coloneqq y_{z_m}(z_n)$, so that $\chi_r Y = \mathbb{1}$. Y is symmetric because $\mathbb{D}$ is, and so is χ_r (Eq. 13, Eq. 14, Eq. 19). Hence $(Y\chi_r)_{nm} = \sum_l \chi_{r,ml} Y_{ln} = (\chi_r Y)_{mn} = \delta_{nm}$, the same absolutely convergent series.
- Let $K(u) \coloneqq \beta H(-u) + \lambda\alpha_k(u)$. Since $\alpha_k(u)\alpha_k(-u) = h^2$ (Eq. 55), for every u

$$\frac{\beta H(-u) - \lambda\alpha_k(u)}{K(u)} + \frac{\beta H(u) - \lambda\alpha_k(-u)}{K(-u)} = \frac{2\beta^2 e(u)}{K(u)K(-u)}. \tag{79}$$

  At a root the right side vanishes, so Z_j and γ_j do not depend on the sign of u_j.
- *Confluent roots.* By Eq. 79, the odd part $[Z(u) - Z(-u)]/2 = \beta^2 u P(u) e(u)/(K(u)K(-u))$ vanishes at the roots, so $Z(s_j) = Z_j$ for the even part $Z(s)$ of Theorem 4. Hence the sum over j in Eq. 33 is $\lambda\rho$ times the divided difference of $s \mapsto Z(s)/((z_n^2 - s)(z_m^2 - s))$ at s_1, s_2. Near a double root $u_\star^2$, $Z(u)$ is analytic in u at $\pm u_\star$: $K(u_\star) \neq 0$, and at a root $K(u) = 0$ iff $K(-u) = 0$ (Eq. 55, Eq. 29). So $Z(s)$, an even analytic function of u, is analytic in s, also for $u_\star = 0$. As μ approaches a value where $s_1 = s_2$ (a zero of the discriminant of e, a nonconstant analytic function of μ), the divided difference converges to the derivative stated in Theorem 4, and Lemma 17 extends $\chi_r Y = \mathbb{1}$.
- For $r = \mathrm{m}$ and $k = 0$ the atomic pair has $F_1 = P(a)/(a^2 - s_2) = 0$ and $W(z; a^2) = 0$ (Appendix A). So its terms in Step 4 vanish, and Step 3 is needed only for the non-atomic pair.
- *Excluded cases.* Steps 1–4 assumed $u_j \neq 0$ and $P(u_j) \neq 0$ for the non-atomic pairs. For $r = \mathrm{m}$, $k = 0$, $P(u_2) = 0$ would mean $s_2 = a^2 = s_1$, which is excluded. Otherwise, fix k: the values of μ where an assumption fails are zeros of $e(0) = s_1 s_2$ or of the resultant of e and P. These are nonconstant analytic functions of μ, so such values are isolated. Eq. 33 has neither u_j nor $P(u_j)$ in a denominator and depends on s_j only. Near such a value, $\chi_{r,nl} = O(|z_l|^{-2})$ for $l \neq n$ and Y_{lm} is bounded, uniformly; so Lemma 17 extends $\chi_r Y = \mathbb{1}$.
- *Lattice roots.* The case $e(z_n) = 0$ is treated in Appendix B. □

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A Rational identities

Each identity is checked exactly with SymPy: (A1)–(A5) by `doped-gamma/verify_identities.py`, (A6) and (A7) by `doped-gamma/verify_dm.py` (both channels, $k \neq 0$ and $k = 0$, all parameters free).

(A1) the twelve terms of Appendix C equal the two-term form in the proof of Lemma 1; and $\Pi(x_n, x_m) = (z + v - 2a)^2/(Q_-(z)Q_-(v))$.

(A2) the z^0 coefficient in the proof of Proposition 15: $4a^2\rho(2a - v)Q_+(v) - 2aq_0Q_+(v) + 4a^2(1 - \rho)q_0(v + 2a) = -2a(v^2 + 2q_0 - 4a^2)(v^2 + 2a\rho v + q_0) + 2aP(v)$.

(A3) Eq. 40; $zv + q_0 = a^2 + (t_n + t_m)/2$ at transfer $w' = b - x_n - x_m$; Eq. 45 and Eq. 46 from the twelve terms of Appendix C at w' with $c = \rho/b$.

(A4) $\Xi = \mathrm{diag}(-1/a, K_\mathrm{s}/(ad_+d_-))$ in the proof of Theorem 3, and $d_+d_- = \mathcal{P}(-A_\mathrm{s})$.

(A5) the δ_{nm} and δ_{k0} identities in the proof of Proposition 15, and $H(a) = 4a^2p_2$, $H(-a) = 4a^2p_0$ for $r = \mathrm{m}$, $k = 0$.

(A6) Step 2 of the proof of Theorem 4: $-\beta H(z)+\lambda\overline{c}+\mathrm{PV}\sum_l N(z,z_l)/P(z_l)=\beta N_2(z)/(2\varepsilon a)$, the sum evaluated by Lemma 8 and Lemma 16.

(A7) Eq. 68, where $\overline{\mathcal{I}}$ is evaluated by Lemma 16 and, with $\varpi := 4(p_0p_2-p_1^2)$,

$$\begin{aligned} W_0(s) &:= \varepsilon a q_0\big[\varpi-2\varepsilon(1-\rho^2)\big](s+q_0+2\varepsilon a^2)+\delta\beta a^4\psi(s),\\ W_2(s) &:= \varepsilon a\varpi(s+q_0+2\varepsilon a^2)-\delta\beta a^2\psi(s),\\ \psi(s) &:= -4\varepsilon p_1(p_0+p_2-\rho^2)\big[s+(1+2\varepsilon)a^2\big]+(1+\varepsilon)\rho^2\varpi(s-a^2). \end{aligned} \tag{80}$$

For $r=\mathrm{m}$ and $k=0$ ($\varepsilon=-1$, $q_0=a^2$) this shows $W_0(a^2)=W_2(a^2)=0$.

B Analytic details

Lemma 17 (continuity). Let $\chi(p)$ and $Y(p)$ be matrices indexed by the lattice, depending on a real parameter p (μ or β) near p_0, with entries continuous at p_0. Suppose that for all n, m there are c_{nm} and a finite set of indices l outside of which $|\chi(p)_{nl}Y(p)_{lm}|\le c_{nm}(1+l^2)^{-1}$ for all these p. If $\chi(p)Y(p)=\mathbb{1}$ for $p\ne p_0$, then also $\chi(p_0)Y(p_0)=\mathbb{1}$.

Proof. For fixed n, m, the finitely many exceptional terms of $\sum_l \chi_{nl}Y_{lm}$ are continuous at p_0, and the sum of the others is continuous by dominated convergence. □

Zeros of $e(z_n)$ at $\mu\ne 0$. Since e has real coefficients and $z_{\overline{n}}=z_n^*$, $e(z_{\overline{n}})=e(z_n)^*$: lattice zeros of e come in mirror orbits $O=\{n,\overline{n}\}$, with one element if $n=\overline{n}$. Let e vanish on one orbit O at $\mu=\mu_0\ne 0$, the other assumptions of Theorem 4 holding (with $s_1\ne s_2$ at μ_0). As $e(z_n)$ is a nonconstant analytic function of μ, Theorem 4 holds for $\mu\ne\mu_0$ near μ_0. Each root u_j that tends to a point z_c, $c\in O$, has $(z_c-u_j)\alpha_j\to\rho\ne 0$ and $K_j\to\infty$, since $(z_c-u)S_k(u)\to 1$ as $u\to z_c$ (Lemma 7). In the column y_v of Step 4 with $v=z_m$, $m\notin O$, the coefficients D_v, E_v and $F_jg_j(v)$ stay bounded. By Eq. 71, for $c\in O$,

$$\frac{\varphi_j(z_c)}{K_j}=\frac{\beta H(-u_j)/(z_c-u_j)+\lambda\alpha_j/(z_c+u_j)}{\beta H(-u_j)+\lambda\alpha_j}\to\frac{\beta H(-z_c)}{\lambda\rho}+\frac{1}{2z_c}\quad(u_j\to z_c), \tag{81}$$

it tends to $1/(z_c+z_{c'})$ if $u_j\to z_{c'}$ with $c'\ne c$, and it stays bounded otherwise. So $Y_{cm}=y_{z_m}(z_c)=Y_{mc}$ converges for $c\in O$, $m\notin O$, boundedly in m. For the block $(Y_{cc'})_{c,c'\in O}$, row $l\notin O$ of $\chi_rY=\mathbb{1}$ gives $\sum_{c\in O}\chi_{r,lc}Y_{cc'}=-\sum_{l'\notin O}\chi_{r,ll'}Y_{l'c'}$, whose right side converges by dominated convergence. The columns $P(z_l)\chi_{r,lc}=\lambda\rho/(z_l+z_c)+N(z_l,z_c)/P(z_c)$, $c\in O$ (Eq. 61), are linearly independent rational functions of z_l, with distinct poles at $-z_c$. So $|O|$ rows $l\notin O$ can be chosen for which the matrix $\big(\chi_{r,lc}(\mu_0)\big)$ is invertible, and the block converges. All entries of Y converge, boundedly off the diagonal, and Lemma 17 gives $\chi_rY=\mathbb{1}$ at μ_0.

Zeros of $e(z_n)$ at $\mu=0$. Let $\beta H(z_n)=\lambda h$ with $n\ne\overline{n}$ at $\beta=\beta_0$, so that $z_n^2=\hat{s}$ and $e(z)=(z^2-\hat{s})(z^2-\hat{s}+2\lambda h/\beta)$. As $\beta\to\beta_0$, $\hat{u}\to z_n$. The diagonal and mirror terms of Eq. 35 contribute $-\lambda hP(z_n)/e(z_n)+O(1)=-\beta P(z_n)/(2(z_n^2-\hat{s}))+O(1)$ to each of the entries (n,n) and $(n,\overline{n})$. Since $S_k(\hat{u})=1/(z_n-\hat{u})+O(1)$, $\hat{D}=2\varepsilon a/(\hat{u}(\hat{u}-z_n))+O(1)$, so the separable term contributes $\beta P(z_n)/(2(z_n^2-\hat{s}))+O(1)$ to the same entries and $O(1)$ to the others. The singular parts cancel.

At a mirror fixed point $n=\overline{n}$ (odd k, $z_n=0$), let $e(0)=0$ at $\beta=\beta_0$. The diagonal and mirror terms of Eq. 35 act on the same entry (n,n). As $\beta^2e(0)=(\beta H(0)-\lambda h)(\beta H(0)+\lambda h)$, they combine to $\beta P(0)/G_2(0)-\beta^2P(0)/(\beta H(0)-\lambda h)$. If $\beta H(0)=-\lambda h$ at β_0, this stays finite, and under the remaining assumptions of Corollary 5 nothing else is singular. If $\beta H(0)=\lambda h$, then $s:=\hat{s}\to 0$, and the combination is $\beta P(0)/s+O(1)$, as $\beta H(0)-\lambda h=-\beta s$. For odd k, $\coth(\beta\hat{u}/2)/\hat{u}=2/(\beta s)+O(1)$, so $\hat{D}=2\varepsilon a/s+O(1)$, and the separable term is $-2\varepsilon a\beta P(\hat{u})\hat{g}(0)^2/\hat{D}=-\beta P(0)/s+O(1)$ at (n,n) and $O(1)$ elsewhere. The singular parts cancel again.

In all these cases Eq. 35 extends continuously, and Lemma 17 (with $p = \beta$) gives $\chi_r Y = \mathbb{1}$ at β_0.

Proof of Corollary 5. Take $\mu \to 0$ at fixed k in Eq. 33. Then $\rho \to 0$, $\overline{c} \to h$ and $C_{nm} \to h\delta_{m\overline{n}}$ (Eq. 16). One root s_j of e tends to $\hat{s}$, the other, s_ℓ, to $-4\varepsilon a^2(p_0 + p_1) - q_0$. At a root u_j put $X_\pm := \beta H(\pm u_j)$. Then $X_+X_- = \lambda^2 h^2$ (Eq. 29) and $X_+ - X_- = 4a\beta\rho u_j$. With $\overline{c}^2 = h^2 + \beta^2\rho^2/4$ (Eq. 4) this gives

$$(X_- - \lambda\overline{c})(X_- + \lambda\overline{c}) = -\rho\big(4a\beta u_j X_- + \lambda^2\beta^2\rho/4\big). \tag{82}$$

By Eq. 30, $K_j = X_- - \lambda\overline{c} + \lambda\rho S_k(u_j)$ and $X_- - \lambda\alpha_k(u_j) = X_- + \lambda\overline{c} - \lambda\rho S_k(u_j)$.

- *The pair j.* Here $X_\pm \to \lambda h$, so the identity gives $(X_- - \lambda\overline{c})/\rho \to -2a\beta\hat{u}$. Hence $K_j/\rho \to \lambda S_k(\hat{u}) - 2a\beta\hat{u} = -2a\hat{u}\hat{D}$, $X_- - \lambda\alpha_k(u_j) \to 2\lambda h$ and $s_j - s_\ell \to 2\lambda h/\beta$. So $\gamma_j \to -2\varepsilon a\beta P(\hat{u})/\hat{D}$ if $\hat{u} \neq 0$.
- *The pair ℓ.* Here $X_\pm \to -\lambda h$, so $K_\ell \to -2\lambda h$, $X_- - \lambda\alpha_k(u_\ell) \to 0$ and $\gamma_\ell \to 0$. (For $r = \mathrm{m}$, $k = 0$ this is the atomic pair, absent anyway.)
- *The dense term.* By Eq. 77, $\lambda\rho\mathbb{D}(z_n, z_m) = \lambda C_{nm}F(z_m) - \lambda\rho z_n \sum_j F_j g_j(z_n) g_j(z_m) \to \lambda h F(z_n)\delta_{m\overline{n}}$, since $z_{\overline{n}} = -z_n$ at $\mu = 0$ and F is even.

So Eq. 33 tends entrywise to Eq. 35. For small $\mu \neq 0$ the assumptions of Theorem 4 hold: $e(z_n) \neq 0$ by continuity (automatically for large $|n|$), $s_1 \neq s_2$, $K_\ell \neq 0$, and $K_j \neq 0$ because $K_j/\rho \to -2a\hat{u}\hat{D} \neq 0$ (for $\hat{u} \neq 0$). The entries of C are bounded uniformly near $\mu = 0$ (by $\beta\rho/(2\pi)$ for $m \neq \overline{n}$, and $\rho/(2\mu) \to \beta p_0$). So $\chi_{r,nl} = O(|z_l|^{-2})$ for $l \neq n$, and the entries of $Y = \Gamma_r/\beta^2 + \boldsymbol{B}^{-1}$, with Γ_r from Eq. 33, are bounded off the diagonal, uniformly near $\mu = 0$. So Lemma 17 gives $\chi_r Y = \mathbb{1}$ at $\mu = 0$; Y stays symmetric. Finally, for even k, $\hat{u} = 0$ occurs at isolated β, where Eq. 35 is continuous in β (as $\hat{D} \neq 0$); Lemma 17 with $p = \beta$ covers it. For odd k, $\hat{u} = 0$ means $e(0) = 0$ at the mirror fixed point, treated above. □

Proof of Proposition 6. Let $\vartheta_n := (1 + |z_n - \mu|^2)^{1/2}$, $\boldsymbol{\vartheta} := \mathrm{diag}(\vartheta_n)$, and $\hat{\chi} := \boldsymbol{\vartheta}\chi_r\boldsymbol{\vartheta}$ on ℓ^2.

- $\hat{\chi} = \kappa_r\mathbb{1} + \hat{\chi}_c$ *with* $\hat{\chi}_c$ *compact*, where $\kappa_\mathrm{d} = \kappa_\mathrm{m} = \beta$ and $\kappa_\mathrm{s} = -\kappa_\mathrm{t} = \beta/2$.
  - $r = \mathrm{d}, \mathrm{m}$: by Eq. 61, $\chi_{r,nn} = -\beta H(z_n)/P(z_n) + O(z_n^{-4})$, so $\vartheta_n^2\chi_{r,nn} \to \beta$. Off the diagonal, $\chi_{r,nm} = O(|z_n|^{-2}\,|z_m|^{-2})$ by Eq. 13, Eq. 14 and Eq. 19. So these entries of $\hat{\chi}$ are square-summable, i.e. Hilbert–Schmidt.
  - $r = \mathrm{s}, \mathrm{t}$: $t_n/\vartheta_n^2 \to 1$, so $\vartheta_n^2 B_n \to -\beta/2$, $\vartheta_n^2 L_\mathrm{s}(t_n) \to \beta/2$ and $\vartheta_n^2 L_\mathrm{t}(t_n) \to -\beta/2$ (Eq. 41). By Eq. 44, $\hat{\chi} - \kappa_r\mathbb{1}$ is a matrix supported on the diagonal and the mirror with entries tending to zero, which is compact, plus (for $r = \mathrm{s}$) $\boldsymbol{\vartheta}\chi_{\mathrm{s,\,reg}}\boldsymbol{\vartheta}$, which is Hilbert–Schmidt by Eq. 45.
- $\hat{\chi}$ *is invertible.* For each m, $\hat{\chi}$ maps $\boldsymbol{\vartheta}^{-1}\chi_r^{-1}\boldsymbol{e}_m$ to $\vartheta_m\boldsymbol{e}_m$: the n-th entry of the image is $\vartheta_n\sum_l \chi_{r,nl}(\chi_r^{-1})_{lm} = \vartheta_n\delta_{nm}$, an absolutely convergent series. The vector $\boldsymbol{\vartheta}^{-1}\chi_r^{-1}\boldsymbol{e}_m$ lies in ℓ^2, because by the explicit formulas $(\chi_r^{-1})_{nm}$ is bounded in n for $n \notin \{m, \overline{m}\}$. So the range of $\hat{\chi}$ contains all basis vectors; being closed (a compact perturbation of $\kappa_r\mathbb{1}$ has closed range), it is all of ℓ^2. A compact perturbation of $\kappa_r\mathbb{1}$ that is onto is also injective (Fredholm alternative), hence invertible.

Convergence. Let $\mathbb{1}_M$ be the projection onto the box. Then $\hat{\chi}_M := \mathbb{1}_M\hat{\chi}\mathbb{1}_M + \kappa_r(\mathbb{1} - \mathbb{1}_M)$ converges to $\hat{\chi}$ in norm, since $(\mathbb{1} - \mathbb{1}_M)\hat{\chi}_c \to 0$ and $\hat{\chi}_c(\mathbb{1} - \mathbb{1}_M) \to 0$ for compact $\hat{\chi}_c$. So $\hat{\chi}_M$ is invertible for all sufficiently large M, and $\hat{\chi}_M^{-1} \to \hat{\chi}^{-1}$ in norm. On the box, $\hat{\chi}_M^{-1} = \boldsymbol{\vartheta}^{-1}(\mathbb{1}_M\chi_r\mathbb{1}_M)^{-1}\boldsymbol{\vartheta}^{-1}$. □

C The regular part

The regular part in Eq. 15 is [Ref. [2], Eq. (D3), at zero field]

$$\begin{aligned} R_{\mathrm{reg}}(x,y) = (p_0+p_1)\Big[&-\frac{1}{(y+w+a)(y+a)(x-a)}+\frac{1}{(x+w-a)(x-a)(y+a)} \\ &+\frac{1}{(x+w+a)(y-a)(x-a)}-\frac{1}{(x+w+a)(x-a)(y+w+a)} \\ &+\frac{1}{(y+a)(x+w+a)(y+w+a)}+\frac{1}{(y+a)(x+w+a)(x+a)} \\ &+\frac{1}{(y-a)(x+a)(y+w-a)}-\frac{1}{(y-a)(x+w+a)(x+a)} \\ &+\frac{1}{(x+w-a)(x+a)(y+w-a)}-\frac{1}{(y+a)(x+w-a)(x+a)}\Big] \\ -(p_1+p_2)\Big[&\frac{1}{(y-a)(x+w-a)(y+w-a)}+\frac{1}{(y-a)(x+w-a)(x-a)}\Big]. \end{aligned} \tag{83}$$

D Numerical checks

All scripts are in `doped-gamma/`.

- **Input and sums:** Eqs. 13–15 agree with an independent exact diagonalization to 1.4×10^{-13}, the compact form Eq. 19 to 1.6×10^{-15}; Eq. 24, Eq. 38, Eq. 55 and Eq. 57 agree with direct lattice sums (`check_formulas.jl`).
- **Vertices:** Eq. 25 and Eq. 27 agree with an independent residue-based BigFloat evaluator (`doped_gamma.jl`) to 3×10^{-15}, Eq. 33 at 256 bits to 10^{-65} (300 entries, $k=0,1,2,-3,5$, $|\mu|\le 2\,|U|$); at $\mu=0$, Eq. 25, Eq. 27 and Eq. 35 agree with `HubbardAtoms.jl` (Eq. 19 of Ref. [1]) to 3×10^{-14} (`check_dm.jl`).
- **Proof of Theorem 4:** Eq. 61 agrees with the twelve-term input to 3×10^{-76} (256 bits); the pole-pair images Eq. 74 agree with direct box sums of the twelve-term χ_r to 10^{-11} (44 cases, both channels, both signs of U; `check_dm.jl`).
- **Finite boxes and exceptional points:** box inverses converge like $1/M$ in all channels (`indep_check.jl`; Section 4.3). The confluent form of Theorem 4, Eq. 35 at $\hat{u}=0$ and the removable limits listed under *Divergences* agree with finite boxes or the package to 10^{-7}; at $\beta H(z_n)=-\lambda h$ the box inverse diverges (`check_exceptional.jl`).